第二章 电势

Electric potential

§1 静电场的环路定理

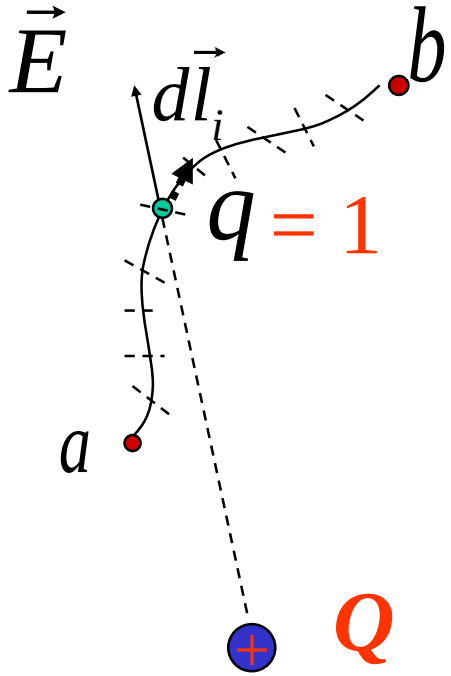
§2 电势和叠加原理

§3 电势梯度

§4 电荷系的静电能

§5 真空中静电场小结

§1 静电场的环路定理



单个电荷电场

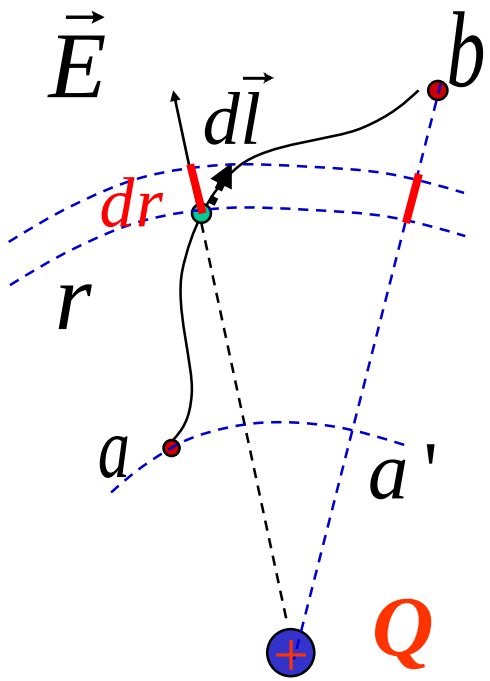
外力克服静电场力作功

$$W = -\int_{(a)}^{(b)} \vec{F} \cdot d\vec{l}$$

$$= -\sum_i \vec{F}_i \cdot d\vec{l}_i$$

电势能的增加 $W(\text{unit}) = -\int_{(a)}^{(b)} \vec{E} \cdot d\vec{l}$

单个电荷电场



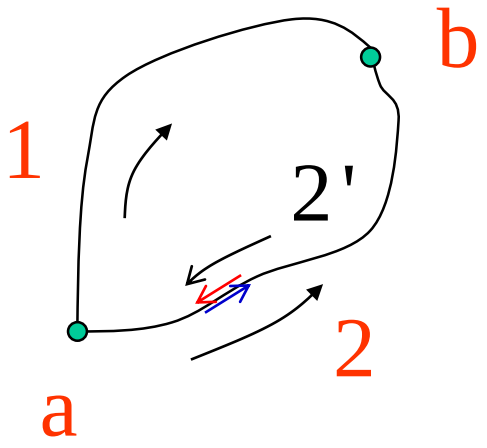
$$-\int_{(a)}^{(b)} \vec{E}\cdot d\vec{l} = -\frac{Q}{4\pi\varepsilon_0}\int_a^b \frac{\hat{r}\cdot d\vec{l}}{r^2}$$

$$= -\frac{Q}{4\pi\varepsilon_0}\int_{a'}^{b} \frac{dr}{r^2}$$

$$-\int_{(a)}^{(b)} \vec{E}\cdot d\vec{l} = -\frac{Q}{4\pi\varepsilon_0}\left(\frac{1}{r_a}-\frac{1}{r_b}\right)$$

只和初末位置有关, 路径无关

多个点电荷场, 利用场强叠加原理知, 结论相同



$$-\int_{(1)} \vec{E}\cdot d\vec{l} = -\int_{(2)} \vec{E}\cdot d\vec{l}$$

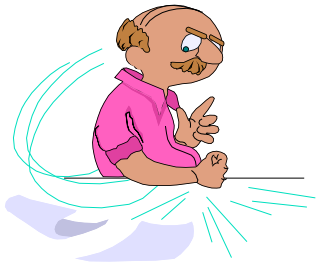
$$-\int_{(1)} \vec{E}\cdot d\vec{l} = \int_{(2')} \vec{E}\cdot d\vec{l}$$

$$\int_{(1)} \vec{E}\cdot d\vec{l} + \int_{(2')} \vec{E}\cdot d\vec{l} = 0$$

静电场环路定理
静电场的保守性

$$\boxed{\oint_L \vec{E}\cdot d\vec{l} = 0}$$

静电场线不能闭合



单选题 1分

设置

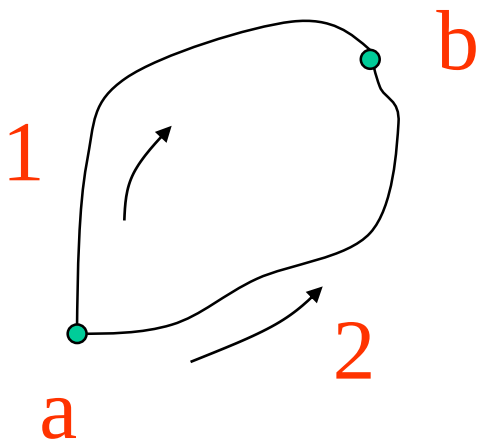
从一点到另一点的电场积分, 总是与路径无关.

A 是

B 否

提交

§2　电势和叠加原理

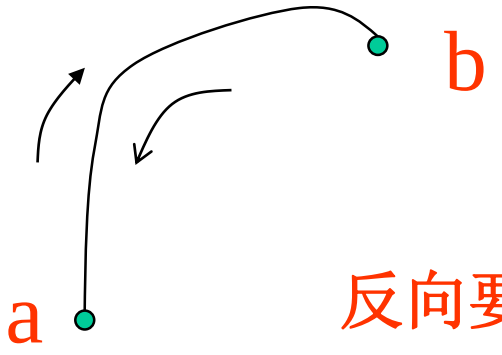


$$\phi(b)-\phi(a)=-\int_{(a)}^{(b)}\vec{E}\cdot d\vec{l}$$

只和初末位置有关, 路径无关

利用这个积分定义函数差 — 电势差

为什么是函数差?



反向要变号

设参考点 $\mathbf{P_0}$ $\qquad \phi(P_0)=0$

电势 $\qquad \boxed{\phi(P)=-\int_{P_0}^{P}\vec{E}\cdot d\vec{l}}$

- 沿电场线电势下降

点电荷电势

$$-\int_{(a)}^{(b)}\vec{E}\cdot d\vec{l}=-\frac{Q}{4\pi\varepsilon_0}\left(\frac{1}{r_a}-\frac{1}{r_b}\right)$$

P_0 选在无限远处

$$\phi(x,y,z) = -\int_{\infty}^{P} \vec{E}\cdot d\vec{l} = \frac{Q}{4\pi\varepsilon_0}\frac{1}{r}$$

电势叠加原理

$$\phi = -\int_{P_0}^{P} \vec{E}\cdot d\vec{l} = -\int_{P_0}^{P} \sum_i \vec{E}_i \cdot d\vec{l}_i = \sum_i \left(-\int_{P_0}^{P} \vec{E}_i \cdot d\vec{l}_i \right) = \sum_i \phi_i$$

点电荷系 $\phi = \sum_i \frac{1}{4\pi\varepsilon_0}\frac{q_i}{r_i}$

单选题 1分

设置

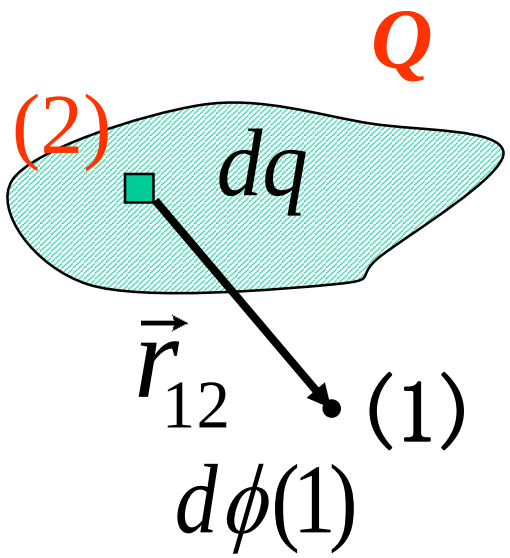
静电场中电势沿电力线上升.

A 是

B 否

提交

若带电体可看作是电荷连续分布的



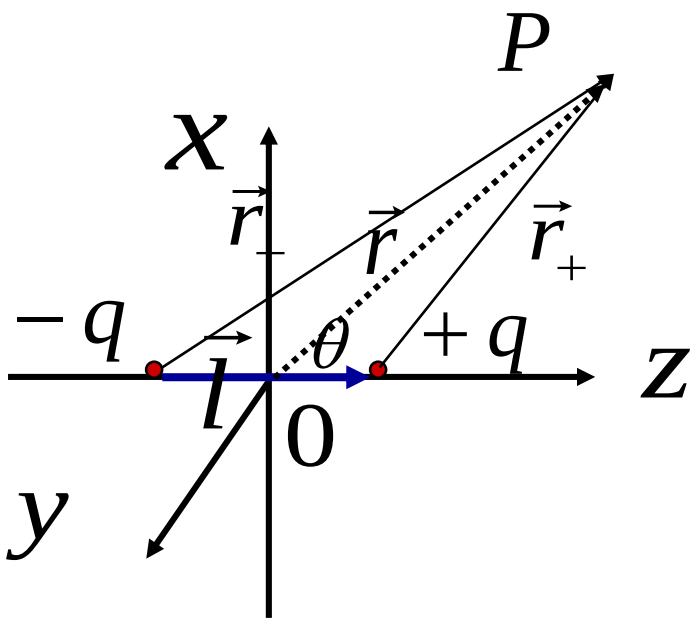
$$\phi(1) = \frac{1}{4\pi\varepsilon_0} \int_{(Q)} \frac{dq}{r_{12}}$$

体电荷分布 $dq = \rho(2)dV_2$

面电荷分布 $dq = \sigma(2)dS_2$

线电荷分布 $dq = \lambda(2)dl_2$

例1 电偶极子势



解： $\phi = \phi_+ + \phi_-$

$$= \frac{q}{4\pi\varepsilon_0}\left(\frac{1}{r_+} - \frac{1}{r_-}\right)$$

$r >> l$

$$\vec{r}_+ = \vec{r} - \frac{\vec{l}}{2} \qquad \vec{r}_+^{\,2} = r^2 + \frac{l^2}{4} - \vec{r}\cdot\vec{l}$$

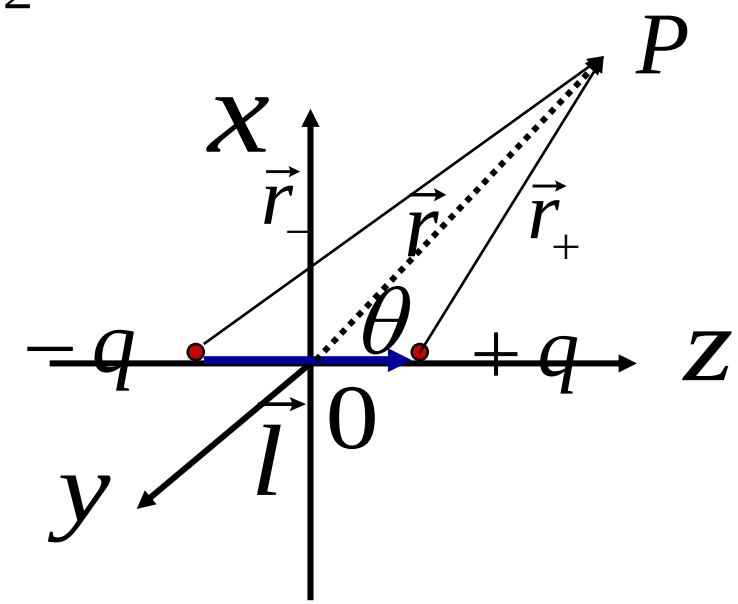
$$= r^2\left(1 + \frac{l^2}{4r^2} - \frac{\vec{r}\cdot\vec{l}}{r^2}\right)$$

$$\vec{r}_- = \vec{r} + \frac{\vec{l}}{2}$$

$$\frac{1}{r_-} \approx \frac{1}{r}\left(1 - \frac{\vec{r}\cdot\vec{l}}{2r^2}\right) \qquad \frac{1}{r_+} \approx \frac{1}{r}\left(1 + \frac{\vec{r}\cdot\vec{l}}{2r^2}\right)$$

$$\phi \approx \frac{q}{4\pi\varepsilon_0}\frac{\vec{r}\cdot\vec{l}}{r^3} = \frac{ql\cos\theta}{4\pi\varepsilon_0 r^2}$$

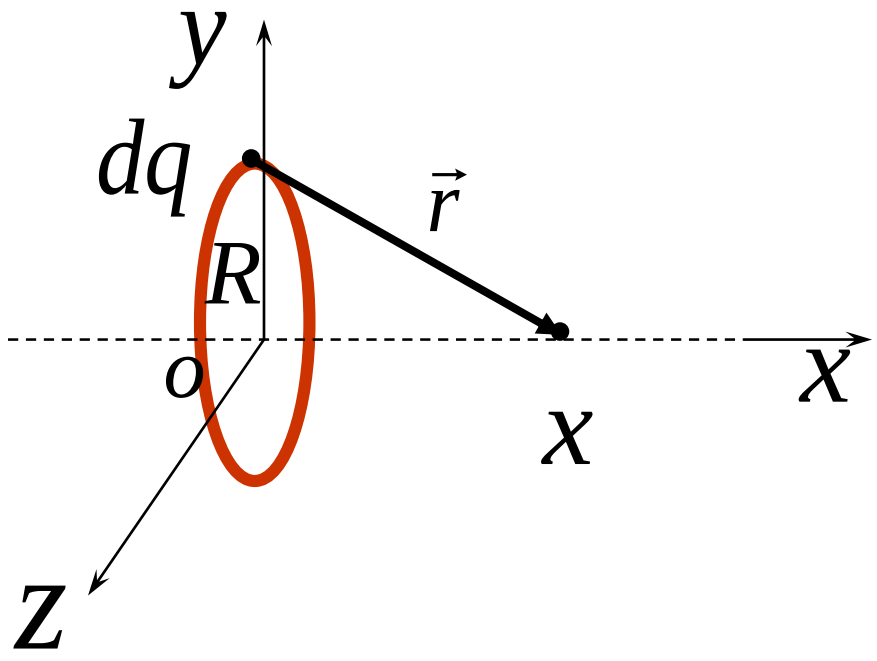
将 $\boxed{\begin{aligned}\vec{p} &= q\vec{l} \\ \cos\theta &= \frac{z}{r}\end{aligned}}$ 代入



$$\phi = \frac{pz}{4\pi\varepsilon_0 (x^2 + y^2 + z^2)^{3/2}}$$

例2　均匀带电圆环轴线上的电势

$Q \quad R \quad x$



解：

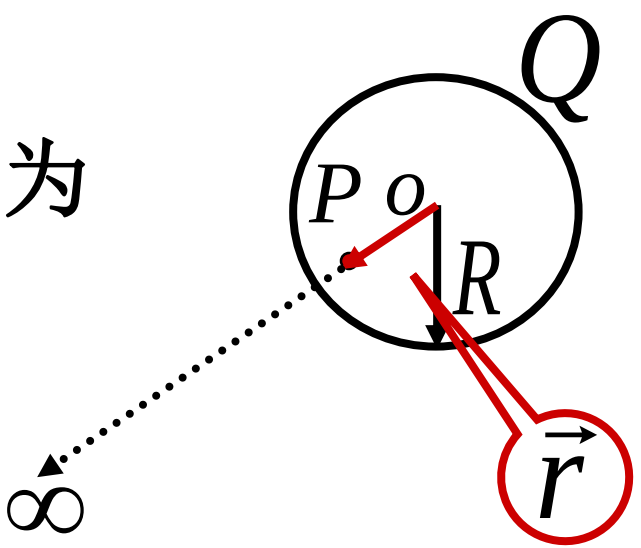
在圆环上任取电荷元 dq

$$d\phi = \frac{dq}{4\pi\varepsilon_0 r}$$

$$\phi = \int \frac{dq}{4\pi\varepsilon_0 r} = \frac{Q}{4\pi\varepsilon_0 r} = \frac{Q}{4\pi\varepsilon_0}\frac{1}{\sqrt{R^2 + x^2}}$$

例3 计算均匀带电球面的电势

解：均匀带电球面电场的分布为



$$\boxed{\begin{array}{ll} r<R & E=0 \\ r>R & \vec{E}=\dfrac{Q}{4\pi\varepsilon_0 r^2}\hat{r} \end{array}}$$

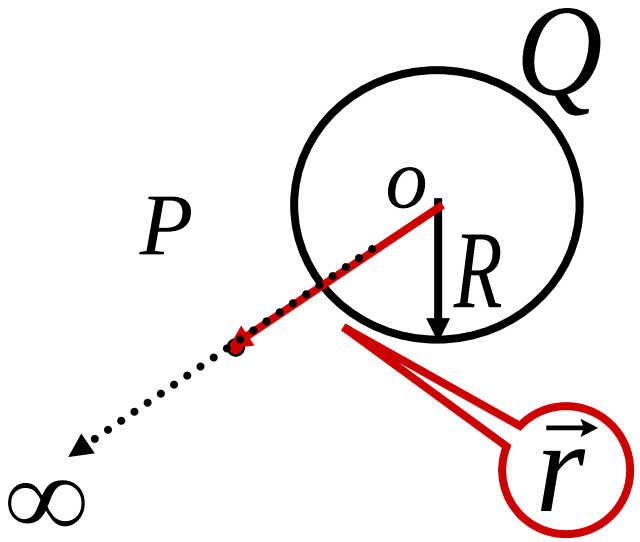
若场点在球内 即 $r<R$ 如图

$$\phi=\int_P^{\infty}\vec{E}\cdot d\vec{l}=\int_r^R 0dl+\int_R^{\infty}\frac{Q}{4\pi\varepsilon_0 r^2}dr=\frac{Q}{4\pi\varepsilon_0 R}$$

等势体

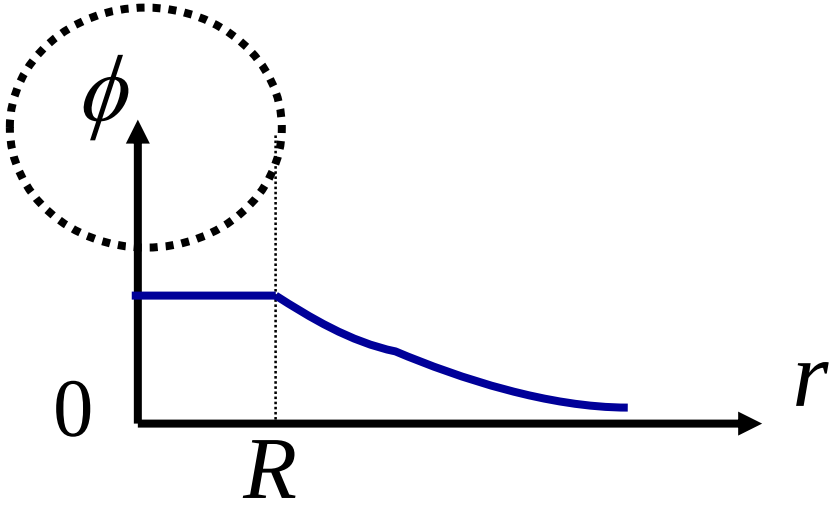
场点在球面外　即 $r > R$

$$\phi = \int_r^{\infty} \frac{Q}{4\pi\varepsilon_0 r^2} dr = \frac{Q}{4\pi\varepsilon_0 r}$$



与电量集中在球心的点电荷的电势分布相同

图示



单选题 1分

设置

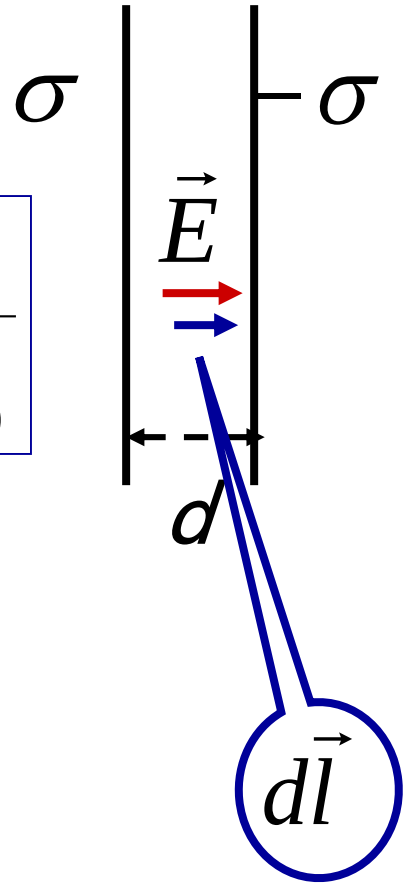
一球面上分布有电荷, 它在球面外产生的电势就好像所有球面上电荷集中于球心时的电势.

A 是

B 否

提交

例4. 平行板电容器两板间的电势差



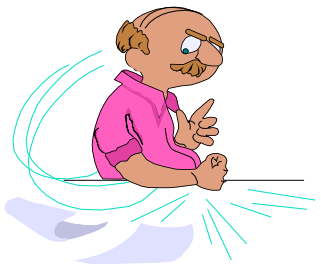
解：

平行板电容器内部的场强为 $\boxed{E = \dfrac{\sigma}{\varepsilon_0}}$

两板间的电势差

$$\Delta\phi = \int_{(+)}^{(-)} \vec{E}\cdot d\vec{l} = \int_{(+)}^{(-)} E dl$$

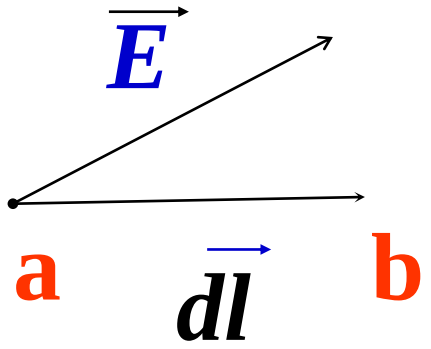
$$= E\int_{(+)}^{(-)} dl = Ed$$



§3　电势梯度

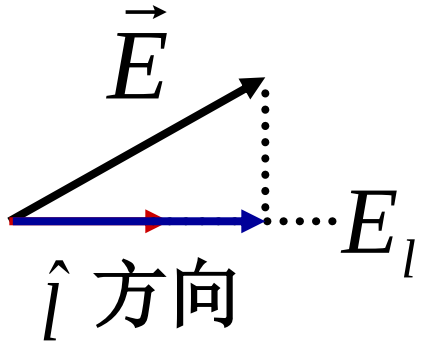
一．电场强度与电势梯度

$$-\int_a^b \vec{E}\cdot d\vec{l} = \phi_b - \phi_a$$



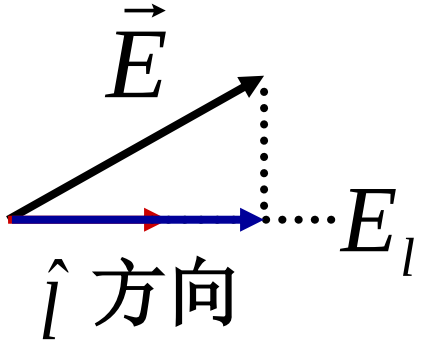
$$-E_l dl = d\phi \qquad E_l = -\frac{d\phi}{dl}$$

$$\boxed{E_l = -\frac{\partial\phi}{\partial l}}$$



即电场强度在 l 方向的分量值

等于电势在 l 方向的方向导数的负值



$$E_l = -\frac{\partial \phi}{\partial l}$$

哪个方向导数最大？

沿电场方向

数学上最大的方向导数叫梯度 ***grad***

结合起来

$$\boxed{\vec{E} = -grad\,\phi}$$

$$\boxed{\vec{E} = E_x\vec{i} + E_y\vec{j} + E_z\vec{k}}$$

$$E_x = -\frac{\partial \phi}{\partial x} \qquad E_y = -\frac{\partial \phi}{\partial y} \qquad E_z = -\frac{\partial \phi}{\partial z}$$

$$\vec{E} = -grad\,\phi = -\frac{\partial \phi}{\partial x}\vec{i} - \frac{\partial \phi}{\partial y}\vec{j} - \frac{\partial \phi}{\partial z}\vec{k}$$

在直角坐标系中

梯度算符

$$\nabla = \frac{\partial}{\partial x}\vec{i} + \frac{\partial}{\partial y}\vec{j} + \frac{\partial}{\partial z}\vec{k}$$

$$\boxed{\vec{E} = -\nabla \phi}$$

$$E_l = -\frac{\partial \phi}{\partial l} = -(\nabla \phi)\cdot \hat{l}$$

单选题 1分

设置

电势降低最快的方向，就是电场方向.

A 是

B 否

提交

例5 利用电势梯度求电偶极子任一点电场强度

$$\phi = \frac{pz}{4\pi\varepsilon_0 (x^2 + y^2 + z^2)^{3/2}}$$

$$E_x = -\frac{\partial \phi}{\partial x} = \frac{pz}{4\pi\varepsilon_0} \frac{3x}{(x^2 + y^2 + z^2)^{5/2}} = \frac{p}{4\pi\varepsilon_0} \frac{3xz}{r^5}$$

$$E_y = -\frac{\partial \phi}{\partial y} = \frac{p}{4\pi\varepsilon_0} \frac{3yz}{r^5}$$

$$E_z = -\frac{\partial \phi}{\partial z} = -\frac{p}{4\pi\varepsilon_0} \left(\frac{1}{r^3} - \frac{3z^2}{r^5}\right)$$

$$\boxed{\vec{E} = E_x \vec{i} + E_y \vec{j} + E_z \vec{k}} \Rightarrow \boxed{\vec{E} = \frac{1}{4\pi\varepsilon_0 r^3}\left(-\vec{p} + (3\vec{p}\cdot\hat{r})\hat{r}\right)}$$

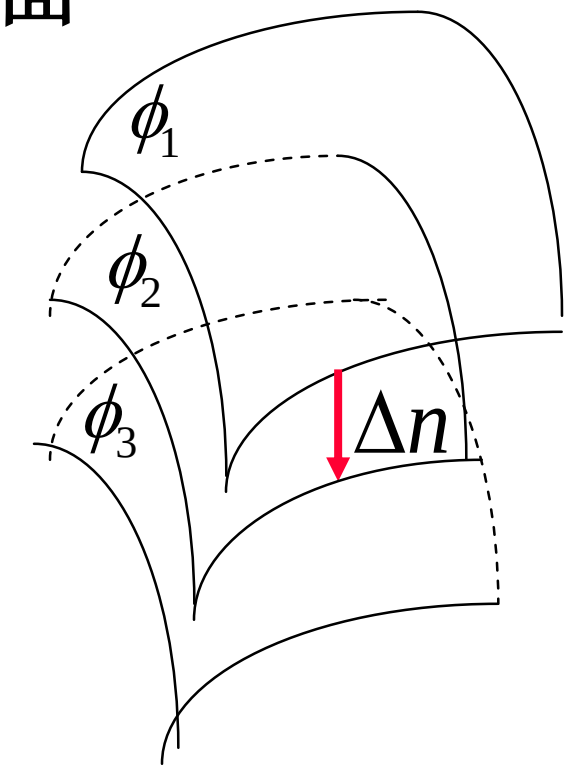
二. 等势面

由电势相等的点组成的面叫等势面

满足方程 $\phi(x,y,z)=C$

$$-E_l dl = d\phi = 0$$

电力线处处垂直等势面



当常量 C 取等间隔数值时

一系列的等势面 $\Delta\phi_{12}=\Delta\phi_{23}$

$\phi_1<\phi_2<\phi_3$

$$\Delta\phi \approx -E\Delta n$$

等势面的疏密反映了场的强弱

单选题 1分

设置

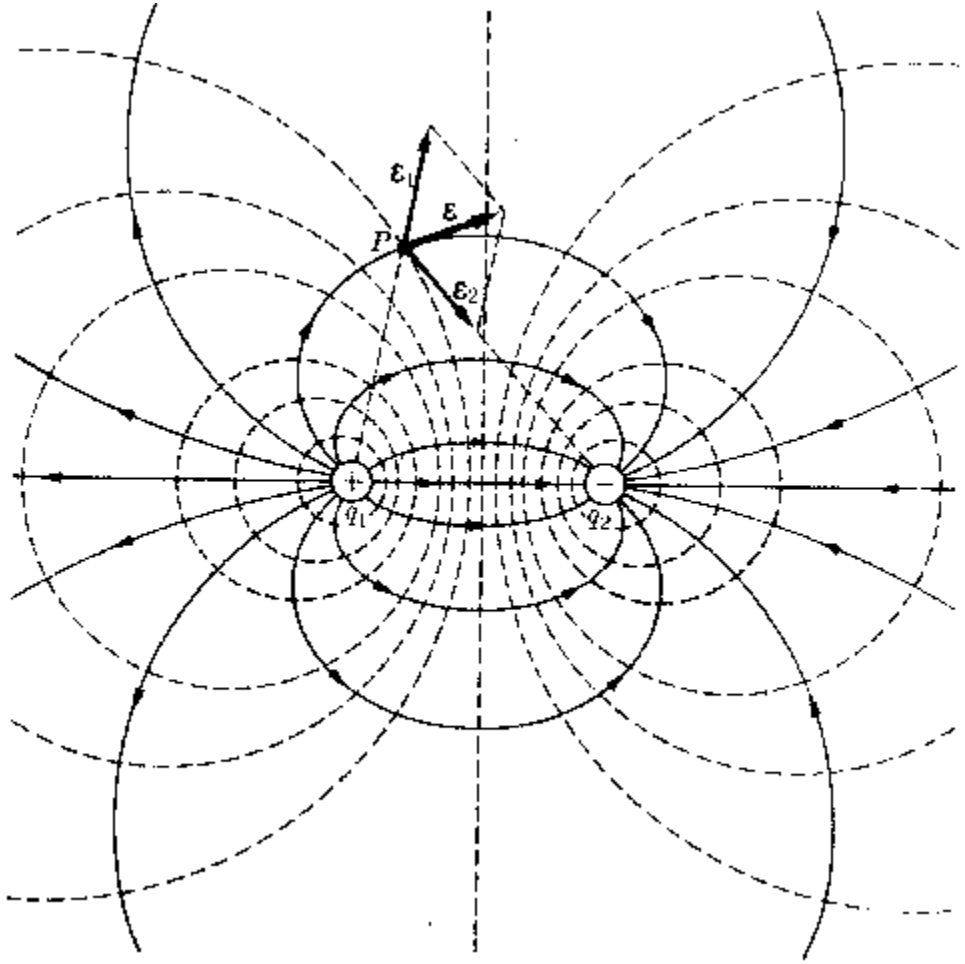
相邻等势面之间电势差值一定，才可以形象表现电场分布.

A 是

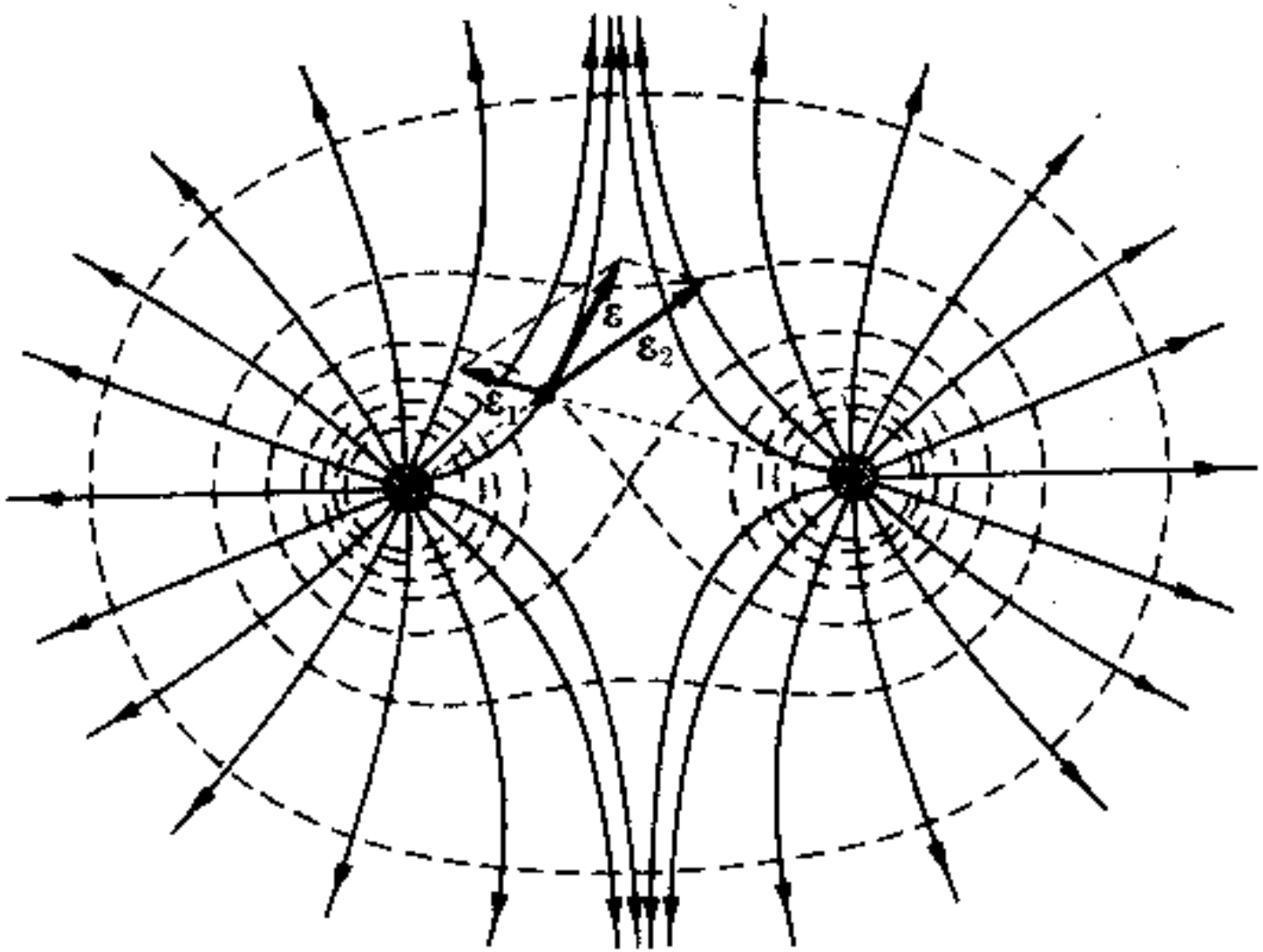
B 否

提交

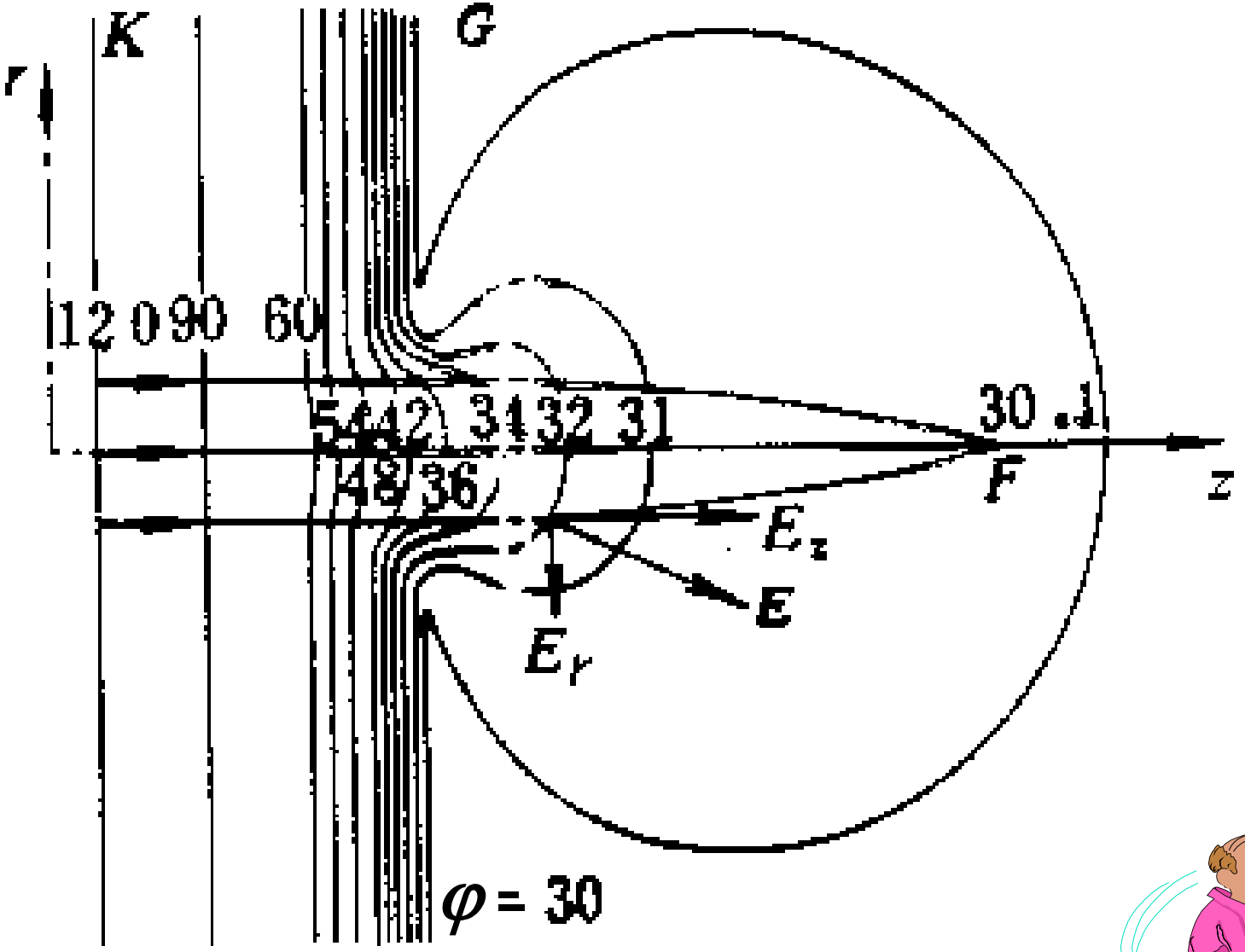
▲某些等势面：

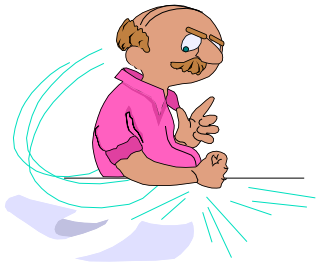


电偶极子的电场线和等势面



两个等量的正电荷的电场线和等势面



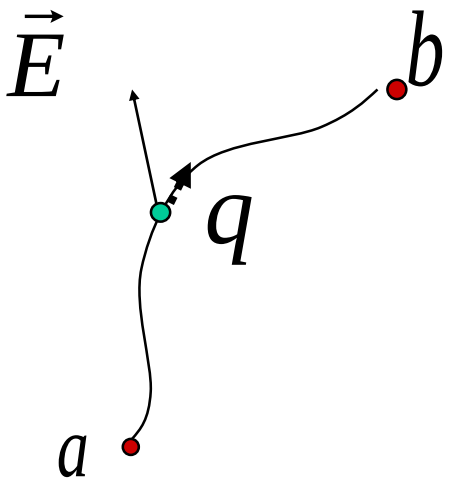


静电透镜的等势面

§4　电荷系的静电能

1. 点电荷在静电场中的电势能

外力克服静电力作的功等于相应电势能的增量



$$A_{ab} = -\int_{(a)}^{(b)} \vec{f} \cdot d\vec{l} = W_b - W_a$$

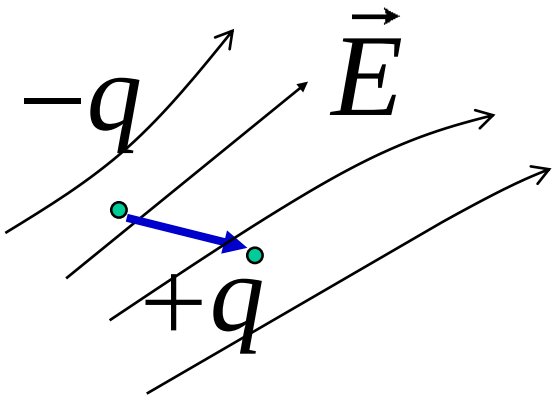
静电场电势能只与位置有关

$$\vec{f} = q\vec{E} \qquad -\int_a^b \vec{E} \cdot d\vec{l} = \phi_b - \phi_a$$

取同一零点　$\boxed{W = q\phi}$

电势能是电荷间的相互作用能

2. 电偶极矩在静电场中电势能



$$W = +q\phi(\vec{r}_+) - q\phi(\vec{r}_-)$$

$$= q[\phi(\vec{r}_+) - \phi(\vec{r}_-)]$$

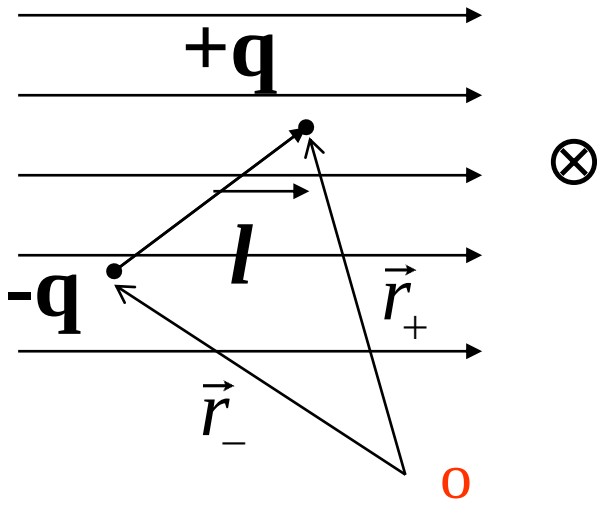
$$= -q\vec{E}\cdot\vec{l} = -\vec{p}\cdot\vec{E}$$

$$\boxed{W = -\vec{p}\cdot\vec{E}}$$

电偶极子作为整体在外场中能量

两个电荷之间相互作用能?　　偶极子自能

另一个角度认识电偶极矩在静电场中电势能:



⊗

在均匀电场中

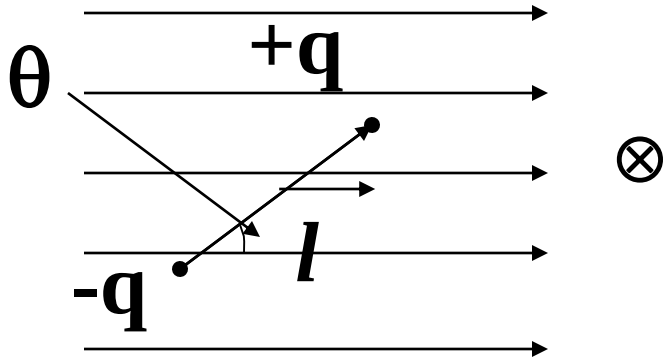
总电场力为零

电场力矩

$$\vec{M} = \vec{r}_+ \times q\vec{E} - \vec{r}_- \times q\vec{E} = q(\vec{r}_+ - \vec{r}_-) \times \vec{E} = \vec{p} \times \vec{E}$$

力矩垂直屏面，电偶极子只在屏面转动

$$\boxed{\vec{M} = \vec{p} \times \vec{E}}$$



* 均匀电场中，
偶极子电势能

转动，电场力矩做功

$$\left|\vec{M}\right| = \left|\vec{p}\times\vec{E}\right| = pE\sin\theta$$

θ=π/2 时电势能为零

$$W_\theta = \int_{\pi/2}^{\theta} M d\theta = \int_{\pi/2}^{\theta} pE\sin\theta d\theta = -pE\cos\theta = -\vec{p}\cdot\vec{E}$$

单选题 1分

设置

电偶极子在电场中只受力矩,不受力.

A 是

B 否

提交

单选题 1分

设置

电偶极子在电场中的能量公式: 负的电偶极矩点积电场. 这里没有考虑两个电荷之间的相互作用能.

A 是

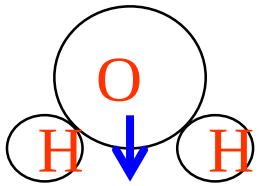
B 否

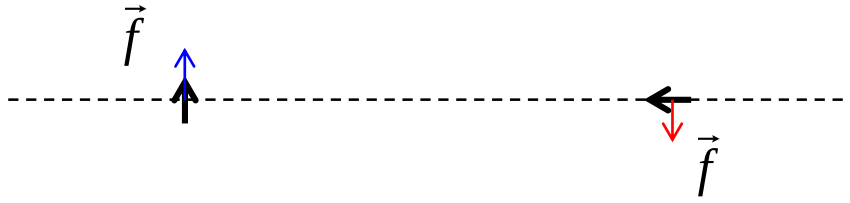
提交

偶极矩间力与牛三?

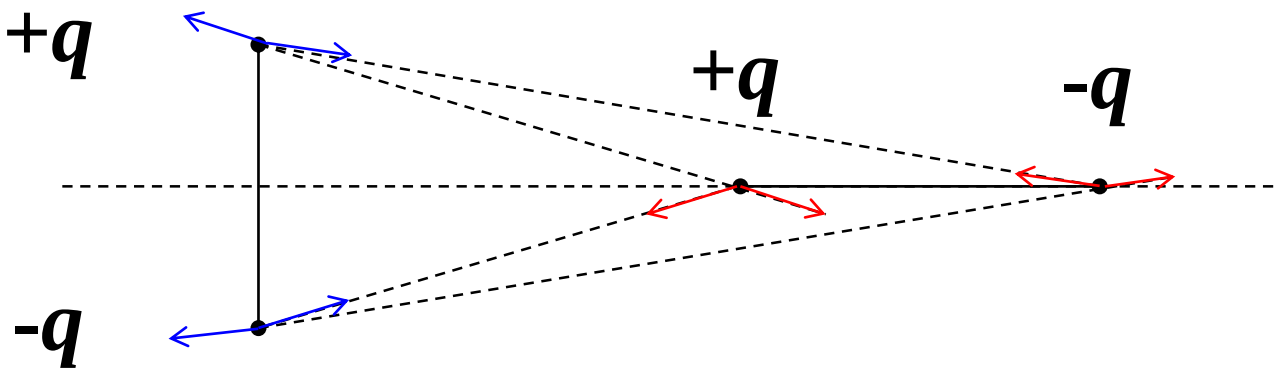
分子力: van der Waals 力

H_2O





$$\vec{M} = \vec{p} \times \vec{E}$$

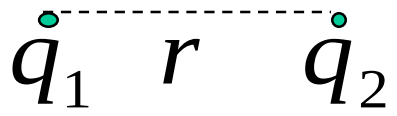


相互作用力大小相等方向相反　总力矩为零

相互作用能?　　$W = -\vec{p} \cdot \vec{E}$

3. 电荷系静电能

两个电荷



想象 q_1 q_2初始时相距无限远

第一步 先把 q_1 摆在某处 外力不作功

第二步 再把 q_2 从无限远移过来 使系统处于状态a 外力克服 q_1 的场作功

$$W = -A_{q_1} = q_2\phi_{21}$$

q_1在q_2所在处的电势

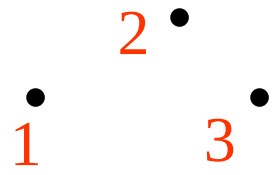
q_2在q_1所在处的电势

也可以先移动 q_2 $W = q_1\phi_{12} = q_2\phi_{21}$

$$W = \frac{1}{2}q_1\phi_{12} + \frac{1}{2}q_2\phi_{21} = \frac{1}{2}q_1\phi_1 + \frac{1}{2}q_2\phi_2$$

$$W = q_2\phi_{21} + q_3(\phi_{31} + \phi_{32})$$

三个电荷:



$$= q_1\phi_{12} + q_3(\phi_{31} + \phi_{32})$$

$$= q_3\phi_{31} + q_2(\phi_{21} + \phi_{23})$$

$$= q_1\phi_{13} + q_2(\phi_{21} + \phi_{23})$$

$$= q_2\phi_{23} + q_1(\phi_{12} + \phi_{13})$$

$$= q_3\phi_{32} + q_1(\phi_{12} + \phi_{13})$$

$$q_1\phi_{12} = q_2\phi_{21}$$

$$q_2\phi_{23} = q_3\phi_{32}$$

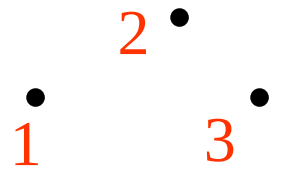
$$q_1\phi_{13} = q_3\phi_{31}$$

$$q_i\phi_{ij} = q_j\phi_{ji}$$

$$W = q_1\phi_{12} + q_2\phi_{23} + q_3\phi_{31}$$

$$W = q_1\phi_{12} + q_2\phi_{23} + q_3\phi_{31}$$

三个电荷:



$$= \frac{1}{2}q_1\phi_{12} + \frac{1}{2}q_2\phi_{21} + \frac{1}{2}q_2\phi_{23} + \frac{1}{2}q_3\phi_{32} + \frac{1}{2}q_3\phi_{31} + \frac{1}{2}q_1\phi_{13}$$

$$= \frac{1}{2}q_1(\phi_{12} + \phi_{13}) + \frac{1}{2}q_2(\phi_{21} + \phi_{23}) + \frac{1}{2}q_3(\phi_{32} + \phi_{31})$$

$$= \frac{1}{2}q_1\phi_1 + \frac{1}{2}q_2\phi_2 + \frac{1}{2}q_3\phi_3$$

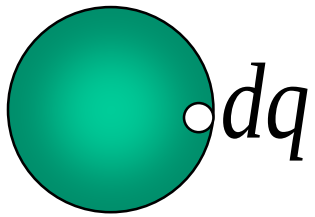
ϕ_i — 除 q_i 以外的电荷在 q_i 处的电势

*N*个电荷:

$$\boxed{W = \frac{1}{2}\sum_i q_i\phi_i}$$

若带电体连续分布

$$\boxed{W=\frac{1}{2}\int_{(Q)} dq\phi}$$



ϕ：所有电荷在dq 处的电势

例6 均匀带电球壳 带电量Q半径r，求电势能

$$W=\frac{1}{2}\int_{(Q)} dq\frac{Q}{4\pi\varepsilon_0 r} \quad =\frac{Q^2}{8\pi\varepsilon_0 r}$$

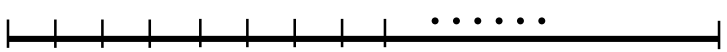
dq 自身处电势？ 均匀带电球电势 $\propto \dfrac{Q}{4\pi\varepsilon_0 r} \sim \rho r^2$

面电荷分布 $\sim \sigma r$

线电荷分布时又如何?

d*q* 自身处电势发散

$$W=\frac{1}{2}\int_{(Q)} dq\phi$$

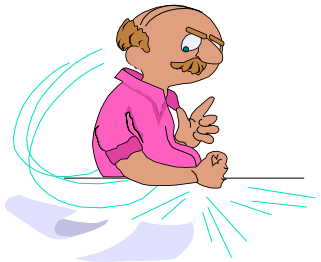


dq 以外电荷在dq 处的电势

线电荷分布电势能仍然发散

点电荷自能发散

$$W=\frac{Q^2}{8\pi\varepsilon_0 r}\xrightarrow{r\to 0}\infty$$



§5　真空中静电场小结

1. 理论框架

库仑定律

+　叠加原理

$$\oiint_S \vec{E}\cdot d\vec{S} = \frac{\sum_i q_{i内}}{\varepsilon_0}$$

$$\oint_L \vec{E}\cdot d\vec{l} = 0$$

$$\nabla\times\vec{E} = 0$$

ϕ

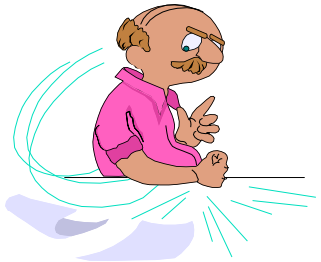
- 静电场范围等价
- 场量描述更具普遍性

2. 计算静电场方法

$$\vec{E} \qquad \phi$$

$$\boxed{\vec{f} = q\vec{E}} \qquad \boxed{W_e = q\phi}$$

- 点电荷公式+叠加原理
- 对称性 + 高斯定理

$$\phi = \int_{(P)}^{P(0)} \vec{E} \cdot d\vec{l}$$

$$\vec{E} = -\nabla \phi$$